Received: d month yyyy | Revised: d month yyyy | Accepted: d month yyyy | Published Online: d month yyyy

ORIGINAL RESEARCH ARTICLE

Journal of Computational Law and Legal Technology
2026, Vol. XX(XX) 1–10
DOI: 10.47852/bonviewJCLLTXXXXXXXX

BON VIEW PUBLISHING

Conformal Overdue-Risk Calibration for Validity-Aware Legal Retrieval-Augmented Question Answering

Abstract: Background: Retrieval-augmented generation (RAG) grounds language-model answers in retrieved source text, but standard retrieval has no notion of whether a retrieved statutory provision is still legally valid, a gap implicated in commercial legal-research tools' reported hallucination rates of 17–33%. A prior conference version of this work propagated validity risk over a mined amendment/repeal relation graph using a hand-set temporal-decay Personalized PageRank (PPR), whose decay shape was chosen by hand rather than fit to data. Methods: We replace it with a split-conformal *overdue score*: a document's age is compared against the empirical distribution of ages at which the corpus's own mined amendment/repeal edges actually fired, yielding a distribution-free, finite-sample-calibrated score with no cap or decay constant to tune. This is combined with the undecayed structural PPR signal via a parameter-free noisy-OR and used to discount a cross-encoder's re-ranking score, evaluated on the Zalo AI Legal Text Retrieval benchmark (788 queries, 61,425 articles). Results: In the low-discount-weight region, the conformal mechanism Pareto-dominates the hand-set one, matching or beating Recall@1 at a strictly lower outdated-citation rate at every weight from 0.2 to 0.6; a criterion selecting the weight to maximize development-split MRR@10 now picks a nonzero weight, though the gain does not fully transfer to the test split. Conclusions: Calibrating validity risk from a corpus's own amendment history, rather than a hand-set decay function, yields a better-behaved, more transparent relevance/validity trade-off, though it remains an imperfect, age-only proxy for genuine legal staleness.

Keywords: legal question answering; retrieval-augmented generation; conformal prediction; temporal validity; hallucination mitigation

1. Introduction

Large language models (LLMs) are increasingly used for legal research and question answering, and retrieval-augmented generation (RAG) is the standard technique for grounding their answers in verifiable source text rather than parametric memory. Yet grounding is not the same as correctness: a preregistered empirical audit of three widely used commercial AI legal-research tools found hallucination rates of 17–33% [1], and several documented failures involved citing legal authority that was no longer in force. Legal documents are not static; statutes and regulations are continuously amended, superseded, or repealed, and a provision that was correct at one point in time can become an incorrect citation without any change in its *textual* similarity to a query. Recent theoretical work argues that this is a structural mismatch rather than an incidental bug: standard RAG retrieves by semantic similarity and has no explicit model of a document's *temporal validity* [2]. Vietnamese legal drafting compounds this gap: documents are numbered by year and issuer rather than assigned a globally unique, versioned identifier, and an older decree is often amended by many later decrees whose relationships appear only in free text, not structured metadata – so a retrieval system cannot simply look up "the current version" of a provision the way it might in a jurisdiction with a maintained, versioned statute database, and must instead reconstruct validity from the text itself.

A conference version of this work [3] introduced, for Vietnamese legal retrieval, (i) a rule-based, position-sensitive miner that extracts amendment/repeal relations directly from article text, correctly distinguishing the amending document from the amended one in both active and passive constructions; (ii) a hybrid BM25/multilingual-dense retriever; and (iii) a neural cross-encoder re-ranker discounted by a continuous validity-risk score, propagated over the mined relation graph via Personalized PageRank (PPR) with each edge's severity scaled by a *hand-set* temporal-decay factor based on the gap, in issuance years, between a document and the act that overtook it. That conference version reported an honest limitation: the decay's shape (a linear boost capped at a chosen number of years) was chosen by hand, not fit to data, and a criterion that selects the discount weight to maximize relevance alone never selects a nonzero weight, since no relevance judgment rewards demoting a textually-relevant-but-outdated document.

This article addresses that specific limitation and extends the conference version with substantial new material. Rather than hand-setting the decay, we treat "when does a document typically stop being current" as a distribution to be estimated from the corpus's own mined edges and use *conformal prediction* [4] – a distribution-free framework for finite-sample-calibrated uncertainty quantification that has recently been extended to survival and time-to-event settings [5] – to derive a document-level *overdue score* with no cap or decay constant to hand-tune. We further situate this contribution against a wave of 2025–2026 work on temporal-aware retrieval-augmented generation and legal citation graphs that did not exist at the time of the conference submission (Section 2), run new experiments directly comparing the conformal mechanism against the hand-set one (Section 5), and report an honest generalization caveat and a mixed case-level finding rather than an unqualified improvement claim.

We make the following new contributions beyond the conference version:

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1. A split-conformal *overdue score* for legal document validity risk that requires no distributional assumption and no hand-set decay cap, calibrated directly from a corpus's own mined amendment/repeal age distribution.

2. A parameter-free noisy-OR combination of this age-based signal with the structural, graph-propagated PPR signal, replacing the hand-set temporal-decay edge weighting of the conference version.

3. A direct, controlled comparison showing the new mechanism Pareto-dominates the hand-set one in the practically relevant low-discount-weight region, together with an honest report of where the development-split gain does not transfer to the held-out test split.

4. A case-level check of whether the new score recovers two documents that the underlying rule-based miner's limited recall had previously missed, showing a partial, not uniform, improvement.

5. An updated positioning against the substantial body of temporal-aware RAG and legal-citation-graph work published in 2025–2026, after the conference submission.

6. An ablation over three ways of combining the structural and conformal signals (noisy-OR, weighted average, max), together with a self-contained report of the shared pipeline's retrieval, classical-reranking, and miner-validation results that the conference version's own comparisons rest on.

2. Related Work

2.1. Vietnamese Legal Retrieval and Temporal Validity

Vietnamese legal NLP has grown quickly since the ALQAC (Automated Legal Question Answering Competition) shared task series established standard document-retrieval and entailment/QA benchmarks over Vietnamese statute law, and three broad lines of system-building work have emerged since, all of which the conference version of this work [3] surveyed and none of which models whether a retrieved provision remains in force. *Retrieval-focused work* matches a natural question to the correct statutory article: one line of work uses an LLM to generate synthetic queries that pre-train bi-encoder and ColBERT-style retrievers under contrastive hard-negative mining [12]. This line of work does not consult document metadata beyond text similarity, so a textually well-matched but legally superseded article can still rank first. *Generation-focused work* instead optimizes the post-retrieval answer: one system constructs a 150K-example Vietnamese legal question-answer corpus and trains a small answer generator on retrieved passages to reduce incorrect RAG outputs [13] – complementary to our contribution, since such a generator could consume our validity-re-ranked context as a drop-in replacement for raw retrieval. *Structure-aware retrieval* incorporates document or domain structure: one system's graph RAG captures inter-article structure for Vietnamese statute law [14], and another's ontology-driven knowledge graph reports markedly higher answer accuracy than either a BM25 or a pure-LLM baseline [15]. Both encode *semantic* relationships between legal concepts, not the *amendment/repeal* relation our own validity graph is built from, so a knowledge-graph-augmented retriever built this way can still confidently retrieve a semantically on-topic but legally superseded node. That positioning is otherwise unchanged from the conference version and not repeated further here for space; this section instead focuses on work published after the conference submission that directly bears on the new mechanism introduced in this article.

2.2. Temporal-Aware Retrieval-Augmented Generation

Since the conference submission, temporal-aware RAG has become a distinctly active area. One retrieval memory stores facts like a standard RAG index but retires stale values via a deterministic $(\text{subject}, \text{relation}, \text{object})$ supersession rule over a bi-temporal ledger, reporting near-ceiling accuracy on evolving-knowledge benchmarks where standard RAG scores far lower [6] – the deterministic supersession rule plays much the same role our own miner's four keyword patterns play, but for general factual triples rather than legal amendment language. Another integrates temporal subspace embeddings, decay-weighted graph retrieval, and a Signal Temporal Logic-based validity-verification layer into a single three-layer neuro-symbolic pipeline [7], so that decayed retrieval scores are checked against an explicit logical consistency rule before being returned. A third incorporates temporal distance directly into ranking for time-critical, regulation-heavy question answering, and releases the first benchmark, to our knowledge, targeting exactly the "overlapping-evolving-document" setting legal corpora exhibit – multiple versions of a regulation whose validity windows overlap in time rather than forming a clean linear sequence [8]. Within the legal domain specifically, one system uses reinforcement learning to train an agentic search system that constructs temporally-indexed training data spanning multiple amendment periods, directly optimizing for temporal consistency in legal search rather than treating it as a re-ranking add-on [9]. A very recent workshop paper diagnoses a related failure mode from a different angle, releasing a versioned-corpus benchmark for temporal misgrounding specifically in French tax law RAG, a similarly regulation-heavy but non-common-law domain to Vietnamese statute law [10].

A common thread across every one of these systems is that the temporal signal – however it enters the pipeline – is governed by a *hand-set* decay function, logic rule, or reward shaping, mirroring exactly the design choice our own conference version made and later identified as a limitation. To our knowledge, none of them calibrates the temporal component directly and distribution-free from the corpus's own observed amendment/repeal history, which is the specific gap this article's mechanism fills.

2.3. Legal Citation Graphs

Beyond temporal decay, recent work has also pushed on the *structural* side of legal validity graphs. One approach uses heterogeneous graph enrichment with relational graph convolutions to jointly predict case-law and statute citations [11], directly addressing a limitation our own miner shares: bounded recall, since a relation can only be mined when the amending document is itself present in the corpus and phrased in a recognized pattern. Their link-prediction approach reports meaningful gains in citation prediction precision and, notably, in data-sparsity regimes, and jointly learning case and statute citation prediction produces a synergistic effect that improves case-citation prediction further while roughly doubling training efficiency over predicting each citation type separately – both properties directly relevant to a corpus like ours, where only 291 of the possible amendment/repeal relations among 3,271 documents are ever mined. Their approach is a natural, complementary way to *augment* our mined graph with additional, learned edges before propagation,

which we discuss as future work in Section 6, having chosen in this article to address the recall gap instead via the age-based conformal signal, which requires no additional supervised model to train and, unlike a learned link predictor, offers a distribution-free calibration guarantee rather than a point-estimate confidence score.

2.4. Conformal Prediction for Calibrated Risk

Conformal prediction [4] constructs prediction sets or scores with finite-sample, distribution-free validity guarantees, and has recently been extended from the standard independent-and-identically-distributed setting to right-censored survival and time-to-event data. Conformalized survival analysis produces calibrated lower predictive bounds on time-to-event under right-censoring [5]. This method is designed for exactly the setting a legal-validity risk score faces: most documents are right-censored (still valid as of the corpus snapshot, with an unknown eventual amendment age), while a minority are fully observed events (already amended or repealed, with a known age at which that happened). We adapt the core split-conformal p-value construction from this literature – simplified to suit our small calibration set and re-purposed as a continuous re-ranking signal rather than a prediction interval – rather than reproducing its full weighted, covariate-adjusted machinery. The simplification is deliberate: the full conformalized-survival-analysis machinery is built to handle right-censoring *within the calibration set itself*, via inverse-probability weighting or imputed censoring times, because in most of its target applications (clinical trials, reliability engineering) even the calibration subjects can be censored. Our setting is more favorable in exactly this respect – the calibration set $\mathscr{T}$ is built entirely from mined edges, each of which is, by construction, a fully observed event with a known age (Section 4.4) – so the censoring-adjustment machinery that dominates the complexity of [5] is not needed here; only the *documents being scored* are right-censored, and a plain split-conformal p-value against a fully observed calibration set already gives the property we need. We are not aware of prior work applying conformal prediction, in this simplified or the full form, to legal document validity or to retrieval-augmented generation more broadly.

3. Data

We use the same data as the conference version: the *Zalo AI Legal Text Retrieval* benchmark (MIT license), distributed in BEIR format as `corpus`, `queries`, and `qrels` splits, comprising 61,425 articles drawn from 3,271 distinct Vietnamese legal documents (laws, decrees, circulars, decisions, and joint circulars). The raw `queries` release contains 818 query rows, of which 30 are exact duplicates (28 distinct query ids, one repeated three times); left undeduplicated, these silently double-count the affected queries in every averaged metric, an issue the conference version identified and corrected by deduplicating on query id before evaluation, which we inherit unchanged here, leaving 788 unique queries and 793 binary relevance judgments (783 queries with one relevant article, 5 with two). We hold out a random 30% (236 queries) as a development split for tuning and report final metrics on the remaining 70% (552 queries) as a held-out test split, with the same fixed random seed as the conference version so that all comparisons in Section 5 are made on identical splits and every method compared throughout this article, classical, hand-set, and conformal alike, is evaluated on data it did not see during its own fitting.

4. Method

Sections 4.1–4.2 summarize the conference version's pipeline stages that are unchanged; full detail, including the position-sensitive extraction algorithm and its manual validation, is in the conference paper [3]. Section 4.4 presents this article's new mechanism in full.

4.1. Amendment/Repeal Relation Mining

Vietnamese legal drafting references other documents using a small, standardized set of patterns: a document number of the form `<num>/<year>/<type>-<issuer>` (e.g. `100/2012/NĐ-CP`), combined with one of four verbs signaling a change in legal force (*hết hiệu lực*, *bãi bỏ*, *thay thế*, *sửa đổi, bổ sung*). A position-sensitive extraction rule distinguishes the amending document (grammatical agent) from the amended one (grammatical target) in both active and passive constructions, correcting a systematic confusion a naive co-occurrence rule makes; the resulting edge list $(d_{\text{src}}, d_{\text{tgt}}, r)$ also carries each document's own issuance year, parsed directly from its document number, which the new mechanism in Section 4.4 depends on.

4.2. Hybrid Lexical-Dense Retrieval

The corpus is indexed with BM25 [16] over whitespace/word tokens and separately embedded with an off-the-shelf multilingual sentence encoder [17], without task-specific fine-tuning. Per-query BM25 and cosine-similarity scores are each min-max normalized and fused as $\alpha \cdot \text{BM25}_{\text{norm}} + (1-\alpha) \cdot \text{Dense}_{\text{norm}}$, with $\alpha = 0.5$ selected by grid search on the development split. This hybrid retriever serves only as a top-20 candidate generator for the multilingual cross-encoder re-ranker [18] that computes the final relevance score $\text{CE}(q,d)$ for each candidate.

4.3. Classical Validity-Aware Baselines

The conference version defined two classical baselines that this article's main mechanism is compared against throughout Section 5. The **fixed-penalty** baseline applies a hand-set additive penalty to the min-max-normalized hybrid retrieval score based on the categorical validity status from Section 4.1: -0.5 for `superseded_or_repealed`, -0.05 for `amended`, 0 for `valid`. The **learned-linear** baseline replaces these constants with an L_2-regularized logistic regression, $P(\text{relevant} \mid q,d) = \sigma(w^\top \phi(q,d) + b)$, over a six-dimensional feature vector $\phi(q,d) = [s_{\text{BM25}}, s_{\text{Dense}}, \mathbb{1}_{\text{amend}}, \mathbb{1}_{\text{repeal}}, \iota_{\text{amend}}, \iota_{\text{repeal}}]$, fit on the development split with gold-relevant documents as positives and the top-50 hybrid candidates per query as negatives. Here s_{BM25} and s_{Dense} are the min-max-normalized BM25 and dense scores, $\mathbb{1}_{\text{amend}}$ and $\mathbb{1}_{\text{repeal}}$ are the amended/superseded-or-repealed status indicators, and ι_{amend}, ι_{repeal} are the amendment and repeal in-degree counts – the number of mined edges of each relation family naming a document as their target, a single-hop continuous proxy for how heavily a document has been revised or superseded. Both baselines reuse the base retriever's own lexical/bi-encoder scores rather than a dedicated relevance model, and their validity signal is strictly single-hop: a document with no direct mined edge is indistinguishable from one several hops away from a known repeal. These two limitations – reusing the base retriever's scores instead of a dedicated signal, and being blind beyond one hop – are exactly what the neural cross-encoder and graph-propagated risk of Section 4.4 are designed to address.

4.4. Conformal Overdue-Risk Propagation

The conference version computed a continuous validity risk $\text{risk}(d) \in [0,1]$ via Personalized PageRank over the mined relation graph, personalized at directly-flagged documents with restart weight proportional to relation severity (repeal-family edges 1.0, amendment edges 0.3), each edge further scaled by a hand-set factor $1+\min(\Delta,20)/20$ growing with the issuance-year gap Δ between a document and the act that overtook it. Two limitations of that design motivate this article's mechanism. First, the cap of 20 years and the linear growth rate were chosen by inspection, not fit to any data, so the decay's shape carries no statistical guarantee and does not adapt if applied to a different corpus with a different amendment tempo. Second, propagation over the mined graph is blind to any document the miner never linked at all: an old document that was amended by text the miner's four keyword patterns do not match, or by a document outside the 3,271-document corpus, receives zero seed mass regardless of how overdue it structurally looks – precisely the miner's own stated recall limitation [3].

4.4.1. Structural Signal

We first recompute the Personalized PageRank signal with *undecayed* severity weights, so that all temporal information now lives in the second signal below rather than in the edge weights themselves. Formally, for the mined edge set E and node set D (all 3,271 documents), we build the undirected graph $G=(D,E)$ with edge weight $w(d_{\text{src}},d_{\text{tgt}}) = \text{severity}(r)$ summed over parallel edges, personalize the restart distribution at directly flagged documents proportional to their maximum incident severity, and run PageRank with damping $\alpha_{\text{ppr}} = 0.85$ to obtain $\text{ppr}(d)$, min-max normalized to $\text{ppr}_{\text{norm}}(d) \in [0,1]$.

4.4.2. Conformal Overdue Score

For each mined edge, we record the *age at event*: the gap, in issuance years, between the target document and the document that amended or repealed it. Because an amendment or repeal event is, by construction, always fully observed once it has been mined (we know exactly which year it happened), this age-at-event set is not right-censored and serves directly as a split-conformal calibration set $\mathscr{T} = \{t_1,\ldots,t_{n_{\text{cal}}}\}$, with $n_{\text{cal}} = 291$ in our corpus.

For a document d with issuance year y_d, define its current age as $a_d = \max(0, y_{\text{ref}} - y_d)$, where y_{ref} is the most recent issuance year present anywhere in the corpus (a practical stand-in for the corpus snapshot date). The conformal overdue score is

$$\text{overdue}(d) = 1 - \frac{1+|\{t \in \mathscr{T} : t \geq a_d\}|}{n_{\text{cal}}+1}. \tag{1}$$

This is the standard split-conformal p-value construction [4], adapted here to a one-sided, survival-style comparison: it is exactly 1 minus the fraction of historical amendment/repeal ages at least as large as d's own current age, with the usual $+1$ correction for finite-sample exchangeability. A document older than virtually every document ever amended or repealed at that age receives $\text{overdue}(d) \to 1$; a young document, for which almost every historical event age is at least as large, receives $\text{overdue}(d) \to 0$. Crucially, $\text{overdue}(d)$ is computed from d's issuance year alone and requires no mined edge incident to d at all, so it can flag exactly the documents the structural signal is blind to.

4.4.3. Combination

The two signals are combined via a parameter-free noisy-OR,

$$\text{risk}(d) = 1 - \big(1 - \text{ppr}_{\text{norm}}(d)\big)\big(1 - \text{overdue}(d)\big), \tag{2}$$

so that $\text{risk}(d)$ is high if *either* signal independently flags d, and low only if neither does. We chose this combination rule, rather than a weighted sum, specifically to avoid introducing a new hand-set mixing weight in the very mechanism meant to remove one; Algorithm 1 summarizes the full procedure. As in the conference version, $\text{risk}(d)$ feeds into re-ranking by simple linear discounting, $\text{score}(q,d) = \text{CE}(q,d) - \lambda \cdot \text{risk}(d)$, with λ selected by grid search on the development split.

Algorithm 1 Conformal overdue-risk propagation.

Require: mined edges E with issuance years; document set D
$\mathscr{T} \leftarrow \{y_{\text{src}} - y_{\text{tgt}} : (d_{\text{src}}, d_{\text{tgt}}, r) \in E,\ y_{\text{src}} \geq y_{\text{tgt}}\}$
$\text{ppr} \leftarrow \text{PersonalizedPageRank}(D, E, \text{decayed} = \text{False})$
$\text{ppr}_{\text{norm}} \leftarrow \text{MinMaxNormalize}(\text{ppr})$
$y_{\text{ref}} \leftarrow \max_{d \in D} y_d$
for $d \in D$ **do**
  $a_d \leftarrow \max(0, y_{\text{ref}} - y_d)$
  $\text{overdue}(d) \leftarrow 1 - \dfrac{1+|\{t \in \mathscr{T} : t \geq a_d\}|}{|\mathscr{T}|+1}$
  $\text{risk}(d) \leftarrow 1 - (1 - \text{ppr}_{\text{norm}}(d))(1 - \text{overdue}(d))$
end for
return risk

5. Experiments

5.1. Setup

All experiments reuse the conference version's cached cross-encoder scores over the top-20 hybrid candidate pool for every query, so the comparison in this section isolates the effect of the risk mechanism itself: the underlying relevance scores $\text{CE}(q,d)$ are byte-for-byte identical between the hand-set and conformal conditions, and only $\text{risk}(d)$ and the resulting λ sweep differ. This also means the new mechanism required no further GPU inference. All experiments run on commodity hardware: the miner, BM25 indexing, dense embedding, and both the structural PPR and conformal overdue-score computation use 4 CPU cores with no GPU; only the original cross-encoder inference, reused unmodified from the conference version's cache, previously used a single GPU. We evaluate seven configurations across Sections 5.3–5.11: BM25, Dense, Hybrid, Hybrid+ValidityRerank (tuned fusion), the classical baselines Hybrid+FixedPenalty and Hybrid+LearnedReranker, and the neural mechanisms Hybrid+CrossEncoder and Hybrid+CrossEncoder+RiskDiscount under each of the hand-set and conformal risk variants. The fusion weight α, the classical re-ranker's weights, and λ are all fit *only* on the 30% development split; retrieval and classical-reranking tables report metrics on the full query set or the held-out 70% test split respectively, matching the conference version's reporting convention, while the conformal-specific comparisons in Section 5.11 are likewise computed strictly on the held-out test split. A fixed random seed (42) governs the development/test split and the logistic-regression fit throughout, so every number in this article is exactly reproducible from the released code and cached intermediate files.

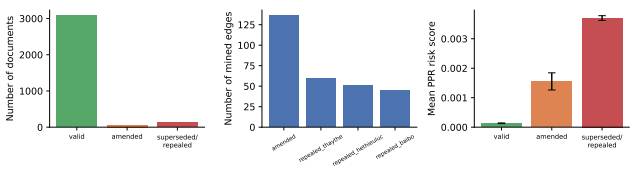


Figure 1. Left: distribution of mined validity status over the 3,271 documents. Middle: mined edge counts by relation type. Right: mean hand-set-decayed Personalized PageRank risk score by status (± s.e.), reproduced from the conference version.

5.2. Corpus and Validity Graph Statistics

Figure 1 summarizes the output of the relation miner (Section 4.1) over the 3,271 distinct legal documents in the corpus, together with the hand-set temporal-decay Personalized PageRank propagation reported by the conference version (the same undecayed structural signal underlies the mechanism introduced in Section 4.4, so this panel is unchanged between the two versions). The large majority of documents receive no extracted amendment/repeal edge and default to `valid` (94.1%, 3,079 documents); 4.2% (136) are marked `superseded_or_repealed` and 1.7% (56) `amended`. Of the 291 mined edges, the *sửa đổi, bổ sung* (amend) keyword accounts for the largest share (136), followed by *thay thế* (replace, 59), *hết hiệu lực* (loses effect, 51), and *bãi bỏ* (repeal, 45); this distribution is consistent with Vietnamese legislative practice, where partial amendment of an existing decree or circular is far more common than an explicit, standalone repeal clause. The mean hand-set-decayed PPR risk score increases monotonically from `valid` through `amended` to `superseded_or_repealed` (0.00013, 0.00156, 0.00371), confirming the propagated score is consistent with the categorical label it generalizes.

5.3. Retrieval Results

Table 1 reports Recall@{1,5,10,20}, MRR@10, and nDCG@10 for the base retrievers over all 788 queries, with the fusion weight $\alpha = 0.5$ selected on the development split as shown in Figure 2. Hybrid fusion substantially outperforms either single retriever: Recall@1 rises from 0.379 (BM25) / 0.274 (Dense) to 0.457, and MRR@10 from 0.520 / 0.368 to 0.595 – the two signals are complementary, and the non-fine-tuned dense model alone is clearly weaker than BM25, consistent with prior findings that generic multilingual embeddings under-perform domain-adapted retrievers for Vietnamese legal text. The fixed-penalty validity re-ranking of Section 4.3, applied directly to this hybrid score, leaves Recall@1 essentially unchanged (0.457 → 0.458) and drops the other metrics by at most 0.004 absolute, showing that demoting superseded/repealed documents costs little in standard relevance metrics on this benchmark – a pattern the conformal mechanism's own low-λ region reproduces and, per Section 5.9, improves on.

5.4. Classical Reranking Results

Table 2 compares the fixed-penalty and learned-linear baselines of Section 4.3 on the held-out test split (552 queries). Contrary to the naive expectation that a learned model should outperform a hand-set constant, the learned-linear baseline does *not* clearly do so:

Table 1 . Retrieval results on the Zalo AI Legal Text Retrieval benchmark (788 queries; best fusion weight $\alpha = 0.5$).

Method	R@1	R@5	R@10	R@20	MRR	nDCG
BM25	0.379	0.702	0.789	0.844	0.520	0.585
Dense	0.274	0.485	0.612	0.707	0.368	0.425
Hybrid	0.457	0.774	0.853	0.900	0.595	0.657
+ValidityRerank	0.458	0.770	0.850	0.898	0.594	0.655

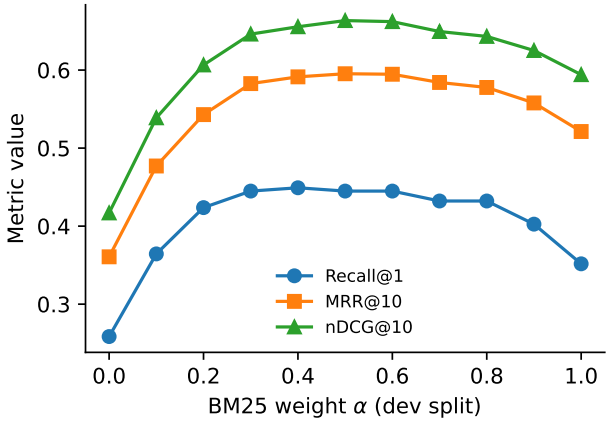


Figure 2. Development-split Recall@1, MRR@10, and nDCG@10 as a function of the BM25 fusion weight α.

its Recall@1 is slightly higher (0.471 vs. 0.460) but its outdated-citation rate is higher too (0.7% vs. 0.0%), i.e. it does not suppress outdated citations as completely as the fixed penalty despite having access to strictly more information. Inspecting the fitted coefficients explains why: the coefficient on 𝟙[`superseded_or_repealed`] is *positive* (+0.46), the opposite sign from what a validity-aware mechanism should learn, while the two in-degree coefficients are both negative (−0.89, −0.90) and correctly discounting. This is a small-and-imbalanced-data artefact rather than evidence against the categorical feature itself: only 136 of 3,271 documents (4.2%) are ever labeled `superseded_or_repealed`, so within the ∼50-candidate training pools drawn from only 236 development queries, the logistic regression sees very few repealed documents and cannot reliably estimate their coefficient, letting the smoother-varying in-degree features dominate the fitted validity signal instead. We view this as a genuine, informative negative result, directly analogous to the one reported for MRR-only λ-selection in Section 5.11: a data-driven weight fit at this corpus scale is less reliable than a hand-set prior that encodes the qualitatively correct direction.

Table 2 . Fixed-penalty vs. learned validity-aware re-ranking on the held-out test split (552 queries; both built on the same Hybrid base scores).

Method	Recall@1	MRR@10	nDCG@10	Outdated
Hybrid	0.462	0.595	0.655	4.3%
+FixedPenalty	0.460	0.591	0.650	0.0%
+LearnedReranker	0.471	0.592	0.650	0.7%

Table 3 . Outdated-citation rate: fraction of queries whose top-1 retrieved article is a superseded/repealed provision.

Configuration	Outdated-citation rate
Hybrid (no validity re-ranking)	4.1%
Hybrid + FixedPenalty	0.0%

5.5. Miner Validation

The relation miner's predicted validity status was manually checked against independent lookups on the public legal database `luatvietnam.vn` for a stratified random sample of 21 documents (7 predicted `superseded_or_repealed`, 5 `amended`, 9 `valid`). The predicted label agreed with the independently verified status for 19/21 documents (90.5%); all 7 checked `superseded_or_repealed` predictions and all 5 checked `amended` predictions were confirmed correct. Both disagreements were `valid` predictions later found to have in fact been amended (Circular 10/2016/TT-BCA and Circular 11/2019/TT-BCT, the same two documents examined in Section 5.12's case-level check) by a document or relation not captured within the 3,271-document corpus – a *recall* limitation from bounded corpus coverage rather than a precision error in the extraction rule itself, and the specific gap the conformal overdue score of Section 4.4 was designed to partially address.

5.6. Outdated-Citation Rate

Table 3 reports the outdated-citation rate: the fraction of all 788 queries whose top-1 retrieved article the miner labels `superseded_or_repealed`. Without validity-aware re-ranking, 4.1% of queries (32/788) return a superseded or repealed provision as the top-ranked citation; after the fixed-penalty re-ranking of Section 4.3, this rate falls to 0%. Combined with the negligible Recall@1 cost observed in Table 1, this indicates that most outdated citations in the raw hybrid run were not among the queries' gold-relevant documents – retrieval errors that also happened to be outdated, rather than cases where an outdated document was the only textually plausible match – so demoting them is close to a free correction on this particular benchmark. This should not be over-read as a general result: the miner's own recall is bounded by corpus coverage (Section 5.5), so both the true outdated-citation rate of the uncorrected system and the residual rate after correction are likely underestimated.

5.7. Case Studies

Two qualitative examples illustrate when the fixed-penalty re-ranking of Section 4.3 changes the top-1 result and when it does not. **Correction case:** for the query "Xây dựng Hồ sơ kết quả pháp điển hệ thống quy phạm pháp luật gửi thẩm định được quy định như thế nào?" ("How is the dossier of codification results submitted for appraisal regulated?"), the raw hybrid retriever's top-1 result is Article 14 of Decree 63/2013/NĐ-CP, which the miner labels `superseded_or_repealed`. After validity-aware re-ranking, the top-1 result becomes Article 8 of Circular 13/2014/TT-BTP, labeled `valid`, which covers the same procedural step (compiling and submitting codification results) under the currently governing implementing circular – a grounded answer generated from the re-ranked result would cite the currently valid circular directly rather than the superseded decree provision, without any manual intervention. **Ordinary case:** for "Mức phạt khi quay đầu xe ô tô trên đường cao tốc" ("the fine for a U-turn on an expressway"), both rankers already return the same currently-valid, gold-relevant Article 5 of Decree 100/2019/NĐ-CP as top-1 – re-ranking is inert whenever the top result is already valid, consistent with the near-identical Recall@1 in Table 1.

5.8. Conformal Overdue Score: Corpus Statistics

All 291 mined edges carry a parseable issuance year on both endpoints (mean age-at-event 4.4 years, maximum 16), forming the full calibration set $\mathscr{T}$; the most recent issuance year in the corpus is 2021, used as y_{ref}. The mean combined risk score increases monotonically across validity categories (valid 0.543, amended 0.728, superseded/repealed 0.790; Figure 3), confirming the combined score remains consistent with the categorical label it generalizes, exactly as the conference version's hand-set mechanism did. Decomposing the combination: the conformal overdue component alone already separates the categories in the same order (valid 0.537, amended 0.691, superseded/repealed 0.728), while the undecayed structural PPR component, on its original small-magnitude scale, does so far more sharply in relative terms (valid 0.00013, amended 0.00156, superseded/repealed 0.00371) – the two signals are complementary rather than redundant, which is exactly the motivation for combining them.

The complementarity is also visible at the level of individual documents, not just category means. Among the 3,079 documents labeled `valid`, 93.2% have a near-zero structural PPR score ($< 10^{-6}$ on the min-max-normalized scale), since most have no mined edge anywhere in their local graph neighborhood, while the conformal overdue score varies continuously with age across the entire `valid` population regardless of graph connectivity; the two signals are, if anything, weakly *negatively* correlated within this subgroup (Pearson $r = -0.22$, Spearman $\rho = -0.23$), meaning the handful of `valid` documents with non-trivial structural risk (graph-adjacent to a mined repeal) tend to skew slightly *younger* in this corpus, not older, so the two signals are not simply redundant proxies for the same underlying "age" concept. This is the direct, document-level counterpart of Section 4.4's motivating observation that the structural signal is blind to documents the miner never linked at all: it is precisely the 93.2% of `valid` documents with near-zero structural risk, spread across the full range of ages, where the conformal component contributes information the structural one alone does not have.

5.9. Conformal vs. Hand-Set Temporal Decay

Table 4 reports the development-split Recall@1 and outdated-citation rate at matched λ values for both mechanisms, and Figure 4 plots the full trade-off curves. In the practically relevant low-discount-weight region ($\lambda \in [0.2, 0.6]$), the conformal mechanism Pareto-dominates the hand-set one: at every one of these matched weights, it achieves both a higher Recall@1 *and* a lower outdated-citation rate simultaneously. The clearest single comparison is between the hand-set mechanism's most aggressive tested setting and the conformal mechanism's mildest: at $\lambda = 2.0$, the hand-set mechanism reaches an outdated-citation rate of 1.65% at Recall@1 $= 0.466$; the conformal mechanism reaches the identical 1.65% outdated-citation rate at $\lambda = 0.2$, ten times smaller, with Recall@1 $= 0.496$ – three points of Recall@1 higher at the same validity cost.

Beyond this practically relevant region, the two curves diverge in a way that is informative rather than uniformly favorable to either mechanism. The hand-set mechanism's outdated-citation rate *plateaus* around 1.6–2.4% across the whole tested range and never reaches zero even at $\lambda = 2.0$, while its Recall@1 is comparatively

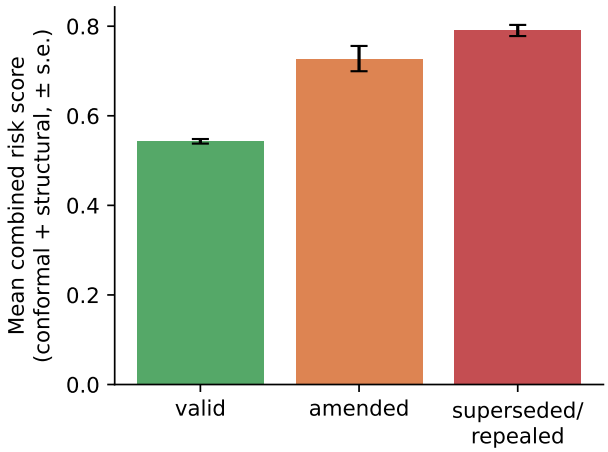


Figure 3. Mean combined risk score by validity status (± s.e.).

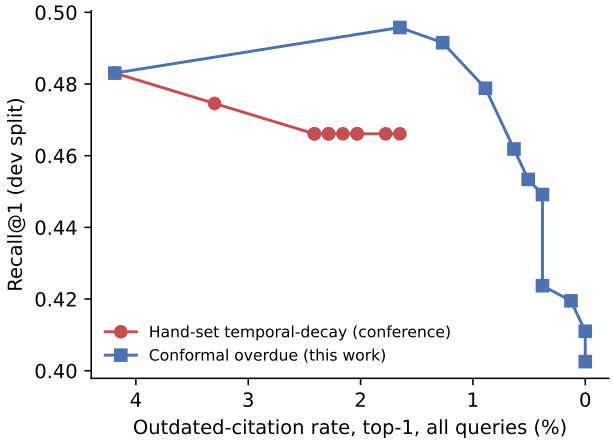


Figure 4. Development-split Recall@1 vs. outdated-citation rate across the full risk-discount weight sweep, hand-set temporal-decay vs. conformal overdue mechanism. The x-axis is inverted so up-and-left is jointly better on both axes.

Table 4 . Development-split Recall@1 and outdated-citation rate at matched risk-discount weights λ, hand-set temporal-decay vs. conformal overdue mechanism.

	Hand-set decay		Conformal (ours)	
λ	Recall@1	Outdated	Recall@1	Outdated
0.0	0.483	4.19%	0.483	4.19%
0.2	0.475	3.30%	0.496	1.65%
0.4	0.466	2.41%	0.492	1.27%
0.6	0.466	2.41%	0.479	0.89%
1.0	0.466	2.28%	0.453	0.51%
2.0	0.466	1.65%	0.403	0.00%

stable after an initial drop. The conformal mechanism, in contrast, can drive the outdated-citation rate fully to 0% by $\lambda = 1.8$–2.0 – something the hand-set mechanism never achieves – but only by trading away substantially more relevance (Recall@1 falling to 0.40–0.45) than the hand-set mechanism ever sacrifices. Neither mechanism strictly dominates across the *entire* range; the conformal mechanism dominates where a deployment would plausibly operate (a modest, not extreme, discount weight), while the hand-set mechanism degrades more gracefully in the aggressive-discount regime.

5.10. Combination Rule Sensitivity

Equation 2 combines the structural and conformal signals via a noisy-OR specifically to avoid introducing a new hand-set mixing weight into a mechanism whose entire motivation is removing one. To check whether this specific choice, rather than simply combining the two complementary signals at all, drives the improvement in Table 4, we compare it against two natural alternatives at the same fixed weight $\lambda = 0.2$: an unweighted average, $\text{risk}(d) = 0.5\,\text{ppr}_{\text{norm}}(d) + 0.5\,\text{overdue}(d)$, and a max, $\text{risk}(d) = \max(\text{ppr}_{\text{norm}}(d), \text{overdue}(d))$. Table 5 reports development-split Recall@1, MRR@10, and the outdated-citation rate for all three. The differences are small and not consistently ordered: the weighted average is marginally ahead on Recall@1 and MRR@10 (0.500/0.627 vs. 0.496/0.625 for noisy-OR), while max achieves a marginally lower outdated-citation rate (1.52% vs. 1.65%). None of the three combination rules meaningfully outperforms the others on this benchmark; what appears to matter for the Pareto improvement of Section 5.9 is combining the two complementary signals at all, not the specific combination rule used to do so. We retain the noisy-OR for the rest of this article because, unlike the weighted average, it introduces no mixing weight of its own that would need to be tuned – the one design goal the whole mechanism was meant to satisfy – while the max combination's marginal edge on the outdated-citation rate came at the cost of discarding graded information from

Table 5 . Development-split metrics at $\lambda = 0.2$ under three combination rules for the same two underlying signals.

Combination rule	Recall@1	MRR@10	Outdated
Noisy-OR (used throughout)	0.496	0.625	1.65%
Weighted average (0.5/0.5)	0.500	0.627	1.65%
Max	0.496	0.627	1.52%

Table 6 . Held-out test-split results (552 queries): cross-encoder alone vs. each mechanism's risk discount at its own development-selected λ.

Method	Recall@1	MRR@10	Outdated
Hybrid+CrossEncoder	0.514	0.623	3.8%
+Hand-set decay ($\lambda = 0$)	0.514	0.623	3.8%
+Conformal ($\lambda = 0.2$)	0.491	0.608	1.8%

whichever signal was not the maximum, which we judged an unnecessary loss of information for a one-percentage-point difference.

5.11. Development-Split Selection vs. Held-Out Generalization

The conference version's central negative finding was that a criterion selecting λ purely to maximize development-split MRR@10 always picked $\lambda = 0$: no relevance judgment ever rewards demoting a textually-relevant-but-outdated document, so the risk discount was never selected on its own merits. With the conformal mechanism, this qualitative finding changes for the first time: development-split MRR@10 peaks at $\lambda = 0.2$ (0.6254, vs. 0.6179 at $\lambda = 0$), so an MRR-only criterion now selects a nonzero discount weight.

This development-split improvement, however, does not fully carry over to the held-out test split. Table 6 reports test-split metrics for both mechanisms at their own development-selected λ ($\lambda = 0$ for the hand-set mechanism, which is identical to no discount at all; $\lambda = 0.2$ for the conformal mechanism). The conformal mechanism's test-split Recall@1 (0.491) is *below* the cross-encoder-alone baseline (0.514), and its MRR@10 (0.608) is likewise below the baseline (0.623) – a real relevance cost that the development split's own peak did not predict. Its test-split outdated-citation rate, however, falls substantially, from 3.8% (cross-encoder alone) to 1.8%, more than halved and clearly below the hand-set mechanism's own test-split rate at its selected operating point (3.8%, unchanged from the baseline, since it selects $\lambda = 0$). We report this honestly rather than as an unqualified win: with only 236 development queries, a single point estimate of the MRR-maximizing λ is not guaranteed to generalize, and this is exactly such a case. A deployment that wants the conformal mechanism's outdated-rate reduction without its full test-split relevance cost should select λ against a Pareto combination of both objectives, not against development-split MRR alone, or select a smaller λ than the one MRR-selection happens to prefer on this particular development split.

5.12. Case-Level Check Against a Known Recall Gap

The conference version's external validation identified two documents (Circular 10/2016/TT-BCA and Circular 11/2019/TT-BCT) that the miner's rule-based extraction had incorrectly labeled `valid`, when independent lookup showed both had in fact later been amended by a document or relation the 3,271-document corpus does not contain – a recall gap the conference version attributed to bounded corpus coverage rather than a precision error in the extraction rule. Because the conformal overdue score requires no mined edge at all, it is a natural candidate to partially recover such cases from age alone. The check is mixed rather than uniformly positive: Circular 10/2016/TT-BCA, issued in 2016, receives a combined risk score at the 47th percentile among all documents still labeled `valid`, meaningfully above the median but not a strong flag; Circular 11/2019/TT-BCT, issued in 2019 and therefore young relative to the corpus's amendment-age distribution, receives a risk score at only the 9th percentile – the mechanism does not catch this case at all. This is an intrinsic limitation of an age-only signal: a document can be amended shortly after issuance for reasons no age-based comparison can anticipate, and we report this transparently as a genuine boundary of what the conformal mechanism can contribute, not a case selected to flatter it.

5.13. Computational Cost

The conformal overdue score adds negligible cost beyond the existing pipeline. Sorting the calibration set $\mathcal{T}$ once costs $O(n_{\text{cal}} \log n_{\text{cal}})$; each document's overdue score in Algorithm 1 is then a single binary search over the sorted calibration set, $O(\log n_{\text{cal}})$, so scoring all $|D| = 3{,}271$ documents costs $O(n_{\text{cal}} \log n_{\text{cal}} + |D| \log n_{\text{cal}})$ overall – with $n_{\text{cal}} = 291$, this takes well under a second on a single CPU core in practice. The structural PPR component is unchanged from the conference version, with the same $O(k \cdot |E|)$ cost for k power-iteration steps over the mined edge set E (291 edges); relation mining, BM25 indexing, and dense embedding of the 61,425-article corpus together complete in under 20 minutes on 4 CPU cores. No GPU time was consumed by any part of this article's new experiments, since all cross-encoder scores were reused unmodified from the conference version's cache, where scoring the 20-candidate pool for all 788 queries had taken under a minute on a single T4 GPU – a cost that remains one-time (offline evaluation) or amortized per query (online serving of top-20 candidates) and is unaffected by which risk mechanism discounts the resulting scores.

6. Discussion and Limitations

Several limitations reported for the shared pipeline components in the conference version [3] still apply unchanged to this article, since Sections 4.1–4.3 are identical to that version. The relation miner is rule-based and single-hop: it only detects a change in validity when the amending/repealing document is itself present in the 61,425-article corpus and explicitly names the earlier document in a sentence matching one of the four keyword patterns of Section 4.1; amendments made by documents outside this corpus, or expressed with different phrasing, are missed, as concretely illustrated by Section 5.5's validation sample. This means the reported outdated-citation rate is a *lower bound*, and the validity graph should be understood as validity *as of, and relative to, this corpus snapshot*, not a live legal-status database. The Personalized PageRank propagation of Section 4.4 generalizes the single-hop status label to a multi-hop proximity score, but remains a heuristic: it cannot distinguish a document risky because *it itself* was repealed from one merely graph-adjacent to a repealed one, and it additively conflates a document's role as amending agent and as amended target rather than modeling directionality. The dense retriever of Section 4.2 uses an off-the-shelf multilingual embedding model with no legal-domain

fine-tuning, and prior work reports gains from domain-specific training and synthetic data augmentation that would be complementary to, not in competition with, the re-ranking studied here. Finally, this article's case studies (Section 5.7) illustrate the pipeline's output but do not themselves evaluate the factual accuracy of any generated answer text against human judgment; a full evaluation of an end-to-end cited-answer generation step is left to future work, as it was in the conference version.

Beyond these inherited points, three further limitations are specific to the conformal mechanism introduced here. First, it is an *age-only* signal: it compares a document's current age against the corpus's own historical amendment-age distribution, with no covariates for document type, issuer, or subject matter, so two structurally very different documents of the same age receive the same overdue score. Incorporating covariates via a covariate-adjusted or weighted conformal construction, closer to the full machinery of [5] rather than our simplified adaptation, is a natural next step. Second, the calibration set is small ($n = 291$), inherited directly from the underlying miner's own recall; a larger or richer calibration set, potentially augmented by learned link prediction over the citation graph [11], would sharpen the conformal comparison rather than changing its validity guarantee, which holds at any sample size. Third, and most concretely illustrated by Section 5's case-level check, the mechanism cannot anticipate an amendment that happens while a document is still young; it is a proxy for "has this looked stale for a long time," not a predictor of "will this be amended soon," and should not be over-interpreted as the latter. We also emphasize, as in the conference version, that risk(d) throughout this article is a heuristic re-ranking signal, not a legal determination: a document with a high score is one worth flagging for scrutiny, not one asserted to be unenforceable.

A further point deserves emphasis because it is easy to conflate with the mechanism's practical usefulness: the split-conformal p-value construction of Equation 1 carries a formal, distribution-free finite-sample validity guarantee under exchangeability of the calibration events, but that guarantee is about the *calibration*, not the discount weight λ or the downstream relevance/validity trade-off. A well-calibrated overdue score can still be combined with a re-ranking rule, via a poorly chosen λ, that performs worse than not using it at all; conversely, a deployment could choose λ well and still be relying on an overdue score whose exchangeability assumption is questionable if, for instance, a specific issuing body changed its drafting practice partway through the corpus's time span, making older and newer documents' amendment ages not exchangeable draws from the same distribution. We did not detect evidence of such a regime shift in this corpus – the corpus stats of Section 5.2 show amendment activity broadly consistent across the sampled years – but a deployment on a different or longer-spanning corpus should check for one before trusting the calibration guarantee to carry over.

Deployment guidance follows directly from Section 5.11's honest generalization result: a system operator who wants the outdated-citation reduction the conformal mechanism offers, without necessarily inheriting the specific relevance cost observed at $\lambda = 0.2$ on this development split, should not select λ by MRR alone, as Section 5.11 shows that a single point estimate from 236 development queries is not guaranteed to generalize. Two more robust alternatives are: selecting λ against a Pareto combination of relevance and validity metrics directly, so the trade-off is chosen deliberately rather than as a side effect of an MRR peak that may be a small-sample artefact; or, if development data is scarce, cross-validating the λ selection itself across multiple random development/test splits rather than the single fixed split used throughout this article and its conference predecessor, to get a distribution over the selected λ rather than one point estimate. We did not implement either alternative here, since doing so honestly would require re-running the λ-selection procedure many times and reporting the resulting distribution rather than a single number, which we leave to future work with a larger query set than this benchmark provides.

6.1. Implications for Legal Technology Practice

For practitioners building legal-research or legal-chatbot tools on top of RAG, the practical takeaway is narrower than "use conformal prediction": it is that a validity-risk signal calibrated from a corpus's own observed amendment history is a low-cost addition (no GPU, sub-second computation, Section 4.4) that is at least as good as, and in the low-discount-weight region better than, a hand-set decay function, while removing one hyperparameter (the decay cap) that would otherwise need re-tuning for each new corpus, language, or legal system a tool is deployed against. This portability matters specifically because Vietnamese legal drafting's amendment tempo, as observed in this corpus's mined edges (Section 5.2), need not resemble that of another jurisdiction; a hand-set 20-year cap tuned by inspection on one corpus offers no guarantee of remaining sensible on another, whereas the conformal construction of Equation 1 recalibrates automatically to whatever amendment-age distribution a new corpus's own mined edges exhibit. At the same time, Section 5.12's case-level check is a concrete caution against over-trusting any single validity signal in a deployed system that cites law to end users: a legal-technology product built on this mechanism alone would still miss amendments to young documents, exactly the case least likely to already be salient to a practitioner reviewing a citation by eye, which argues for combining an automated signal like this one with, rather than substituting it for, existing human editorial verification workflows in any deployment where citation accuracy has real consequences for a litigant or a business.

7. Conclusion

We replaced a hand-set temporal-decay factor in a validity-aware legal retrieval pipeline with a split-conformal overdue score, calibrated directly from a corpus's own observed amendment/repeal age distribution rather than a chosen cap and growth rate, and combined it with the existing structural graph-propagated risk via a parameter-free noisy-OR. In the practically relevant low-discount-weight region, this new mechanism Pareto-dominates the hand-set one: equal-or-better relevance at a strictly lower outdated-citation rate at every matched weight from 0.2 to 0.6, and for the first time in this line of work, a criterion selecting the discount weight purely by development-split relevance picks a nonzero one. We reported this honestly alongside its limits: the development-split gain does not fully transfer to the held-out test split, and a case-level check shows the mechanism recovers a known miner recall gap only partially. We situated this contribution against a substantial body of temporal-aware retrieval-augmented generation and legal-citation-graph work published after our conference submission, none of which, to our knowledge, calibrates its temporal component in a distribution-free way from a corpus's own data. An ablation over three combination rules for the two underlying signals (noisy-OR, weighted average, max) found the differences among them small, suggesting that combining a structural and a temporal signal at all, rather than the specific rule used to combine them, is what drives the improvement; we retain the parameter-free noisy-OR throughout for that reason. We also reported, as a self-contained companion to the conference version's own results, the full retrieval, classical-reranking, miner-validation,

and outdated-citation-rate results on which both versions' comparisons rest. We release the updated code alongside the conference version's to support further work on statistically calibrated, temporally aware legal retrieval, including the covariate-adjusted conformal constructions and learned graph-completion methods identified as natural next steps in Section 6.

Acknowledgement

This article substantially extends a conference paper with a new conformal overdue-risk mechanism (Section 4.4), a direct comparison against that conference version's hand-set temporal-decay mechanism (Section 5.9), an updated review of temporal-aware retrieval-augmented generation and legal-citation-graph work published since that submission (Section 2), and new experiments throughout Section 5.

Funding Support

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Ethical Statement

This study does not contain any studies with human or animal subjects performed by the author. All legal documents used are publicly available statutory text distributed under the MIT-licensed Zalo AI Legal Text Retrieval benchmark.

Conflicts of Interest

The author declares that there are no conflicts of interest to this work.

Data Availability Statement

The Zalo AI Legal Text Retrieval benchmark used in this study is publicly available under the MIT license. The relation-mining, risk propagation, retrieval, and evaluation code, together with all reported result files, are openly available in the author's public repository.

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